📄 REPORT CONTENT (Assignment 2)

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1. Introduction

This assignment focuses on performing sentiment analysis on tweets related to Shah Rukh Khan.
The objective is to classify tweets into positive, neutral, and negative sentiments using machine learning techniques.

This topic was selected because Shah Rukh Khan is a globally popular celebrity, and a large number of opinions are expressed about him on social media platforms.

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2. Data Collection & Preprocessing

A total of **100 tweets** related to Shah Rukh Khan were collected manually from social media platforms.

Preprocessing Steps:

- Converted all text to lowercase
- Removed unnecessary symbols and punctuation
- Cleaned the text for better model performance

Each tweet was **manually labeled** into:

- Positive
- Neutral
- Negative

This labeling helps the model learn sentiment patterns effectively.

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3. Data Splitting

The dataset was divided into:

- **80 tweets (80%) → Training set**
- **20 tweets (20%) → Testing set**

This split was chosen to ensure that the model gets enough data for learning while still having unseen data for evaluation.

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4. Model Training & Classification

Three machine learning classifiers were used:

- Naïve Bayes
- Logistic Regression
- Support Vector Machine (SVM)

The tweets were converted into numerical form using **Count Vectorization**.

Each model was trained on the training dataset and tested on the testing dataset.

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5. Model Evaluation

The performance of each model was evaluated using **Precision** and **Recall**.

Results:

Model	Precision	Recall
Naïve Bayes	0.73	0.65
Logistic Regression	0.60	0.60
SVM	0.56	0.55

Analysis:

- Naïve Bayes achieved the highest precision and recall
- Logistic Regression showed moderate performance
- SVM had the lowest performance

Conclusion:

Naïve Bayes performed better than the other classifiers for this dataset.

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6. Results & Screenshots

- Dataset with labeled tweets was created
- Models were trained successfully
- Precision and recall values were generated

(Screenshots of output and dataset are included in the repository)

Final Conclusion

This assignment demonstrates how sentiment analysis can be performed using machine learning models. Among the tested models, **Naïve Bayes provided the best results**, making it suitable for this type of text classification task.